

Surveillance Awareness & Behavioral Change
The Psychology of Surveillance and Behaviors
Cyber Psychology Portfolio Project
By Jessica

1. Executive Summary

Being watched changes behavior even when the watcher never imposes a single consequence. This report examines how the mirror awareness of digital surveillance, not surveillance itself, and not any actual enforcement action, reshapes what people say, search for, and are willing to communicate online and at work. The evidence for this effect is not anecdotal; It has been measured at the population scale.

The central finding in the surveillance awareness produces two distinct outcomes that are easy to conflate: genuine behavior change (people become more careful because they understand and accept a real risk) and compliance theater (People learn to look compliant while continuing the same underlying behavior through workarounds). Organizations that roll out monitoring tools without accounting for this distinction of the mistakes, a drop in visible activity, and a drop in actual risk when what they are frequently measuring is simply a drop in candor.

2. Background

In 2016, Citizen Lab Researcher Jonathan Penney published the first large-scale empirical study demonstrating a measurable “chilling effect” from government surveillance. Penney analyzed Wikipedia traffic to articles on topics associated with terrorism and government watch list subjects, like specific extremist organizations and their methods, before and after Edwin Snowden’s June 2013 revelations about NSA/PRISM mass surveillance.

The study found a statistically significant, immediate drop in traffic to those specific, privacy- sensitive articles following the widespread publicity around the surveillance revelations, along with a long-term downward shift in the overall trend line for the same content. Traffic to comparable, non-sensitive Wikipedia articles did not show the same pattern. Critically, nothing about the underlying legality of reading a Wikipedia article on a terrorist organization changes. What changed was people's belief that doing so might be observed, logged, and later scrutinized, and that belief alone was sufficient to alter behavior at scale.

The Wikipedia study is significant for this report because it cleanly isolates the psychological variables. There was no new law, no new consequence, and no individual enforcement action against any of the affected readers. The entire effect ran through perception.

3. Psychology Framework

3.1 Chilling Effect Theory

A chilling effect occurs when people restrain lawful, permitted behavior because they fear that behavior might be observed and recorded for later use against them, even in the absence of any actual rule against it. Penney works the extended chilling effect theory, previously mostly a legal concept, into an empirically measurable behavioral phenomenon: self-censorship driven purely by belief and potential observation.

3.2 The Panopticon Effect

Drawn from Bentham's prison design and Foucault's analysis of it, the panopticon effect describes how the mirror possibility of being watched, rather than continuous observation, is sufficient to produce self-discipline. A worker does not need to know a monitoring tool is actively reviewing the message right now; knowing it could be is enough to change what they type.

3.3 Self-Presentation and Front-Stage Behavior

Goffman's dramaturgical model distinguishes "front stage" behavior, performed for an audience, from "backstage" behavior, where a person can be candid without managing impressions. Digital surveillance collapses previously backstage spaces, internal chat, casual email, and personal search history into front-stage performance where every message is composed with an implicit audience in mind.

3.4 Psychological Reactance

Not everyone responds to surveillance with quiet compliance. Reactance theory predicts that when people perceive a threat to their autonomy, a subset will respond by reasserting the restricted behavior through an alternative channel, which helps explain why visible monitoring sometimes coexists with workarounds (personal devices, off-platform chats) rather than with an actual reduction in the underlying activity.

3.5 Compliance vs. Internalization

Kelman's classic model of social influence distinguishes among compliance (behaving correctly only when observed), identification (behaving correctly to maintain a relationship or role), and internalization (behaving correctly because the underlying value is generally accepted). Surveillance-driven behavior change is disproportionately compliance-level: did the case for reroute the moment the perceived observation is absent, which is why it should not be mistaken for improved security culture.

4. Fictional Application: Elite Healthcare

Applying this framework to elite healthcare shows how a well-intentioned monitoring rollout can suppress the very communication and organizational needs, especially candid early reporting of problems, while leaving the underlying risk largely unchanged.

Scenario

Following a data handling incident, elite healthcare deploys activity logging and data loss prevention (DLP) software across its administrative and clinical support staff, with an announcement that “all system activity may be reviewed.” And the weeks that follow, internal chat volume in the general and IT health channels drops sharply. Managers initially read this as a sign that staff have become more careful and disciplined.

In reality, staff have shifted candid conversations to personal phones, flagged a confusing new charting workflow, admitted uncertainty about a documentation requirement, vented about short staffing, and used a group chat outside the monetary system. formal incident and near miss reports, which require staff to describe exactly the kind of mistake the DLP tool might flag, also the client, not because mistakes stop happening, but because employees have become reluctant to put a description of their own era into the monitored log system.

5. Behavioral Analysis

The table below compares staff communication behavior before and after the monitoring rollout, identifying which changes reflect genuine risk reluctance versus displays or suppression of behavior.

Behavior Before Monitoring	Behavior After Monitoring	Psychology Driver
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Casual questions are asked only in the team chat	Same questions as in an off-platform group chat	Reluctance behavior was displaced, not reduced
Nearest airports filed candidly	Near miss reports declined sharply	chilling effect on self-incriminating disclosure
Informal venting about workload and Slack	Venting stops appearing in any monitored channel	front stay self-presentation (Goffman)
Staff Acts IT for help without hesitation	Staff hesitate pre-edit and avoid asking since the questions	Panopticon effect possibly a review, not certainty
Managing to see your messages soon means your problems	The manager cannot distinguish real improvement from suppression	Compliance mistake for internalization (Kelman)

6. Recommendations

6.1 Individual-Level

- Recognize that a felt sense of being watchable change would feel safe to say, even when no rule has actually changed, and factor that awareness into how monitoring announcements are worded and delivered.
- Continue using official channels for safety-relevant disclosure errors, such as message confusion about a process, rather than routing them to unmonitored personal channels where the organization cannot act on them at all.

6.2 Organizational-Level

- Separate DLP and activity monitoring scope explicitly from any review of informal communication content, and communication that separation clearly and specifically vague all activity may be reviewed language maximizes the chilling effect without necessarily improving actual detection.
- Preserve or create a distinct, protected channel for incident and near-miss reporting that staff understands is not subject to the same monetary or disciplinary review, since this is precisely the disclosure most likely to be suppressed.

- Measure program effectiveness using indicators of actual risk (audit findings, breach indicators, and team results) rather than proximity metrics like message volume or reported incidents, which can fall for reasons unrelated to genuine improvement.
- Pilot and communicate monitoring changes gradually with employee input, since reactant-driven workaround behavior is strongest when a policy is perceived as imposed rather than explained.

7. Conclusion

Penney's Wikipedia study demonstrated that surveillance changes behavior through belief alone, without a single enforcement action. The same mechanisms operate inside organizations: a monitoring rollout can produce a visible drop in risky-looking activity. At the same time, the underlying behavior simply relocates to a channel that the organization can no longer see or act on. The Response is not to abandon monitoring, but to design and communicate it in a way that narrows the chilling effect to the behavior actually being deterred, while protecting the candid, self-incriminating disclosure of near misses, honest errors, and early warnings that organizations most depend on.